Stroke

- Observation
  - Posture – Posture of UL and LL should be checked
    - Facial expression should be checked
    - " symmetry " " "
  - Skin – Any superficial lesion / scar
    - Any oedema
    - Patient's behaviour
    - General condition of patient.
- Examination
  - Cranial N examination
    - Facial sensation – CN V
    - " movt – CN V VII
    - Auditory function – CN VIII
    - Motor function of
      - Lips
      - Mouth
      - Tongue – CN 9, 10, 12
      - Palate
      - Pharynx
      - Larynx
- Sensory examination –
  - Specific localized areas of dysfunction are common c̄ cortical lesions and diffuse involvement throughout one side of body suggest deeper lesion involving thalamus and adjacent str.

- Superficial
  - Touch
  - Pressure
  - Temperature
- Deep
  - Proprioception
  - Kinesthesia
  - vibration
- Cortical
  - Stereognosis
  - Tactile localization
  - Two point discrimination
  - Texture recognition

- Motor examination

① Tone - Hypotonicity (Flaccid) tnt immediately after stroke due to effect of cerebral shock.

↓

Short lived but may persist in some patients c̄ lesions of Primary motor cortex or cerebellum.

↓

In 90 % cases, Hypertonicity (spastic) emerges

↓

This conversion of Flaccid → spastic
c/a obligatory synergy

U/E spasticity is strong in

- Scapular retractors
- Sh. add and int rotators
- elbow flexors
- forearm pronators
- wrist and finger flexor

L/E Spasticity is strong in.

- Pelvic retractors
- Hip add and int. rotators
- " and knee ext.
- Plantar flexors
- Toe flexors

- Tightly fisted hand c̄ elbow flexed and held tightly against the chest or a stiff extended knee c̄ plantarflexed foot is seen in stroke.

- Severity of spasticity can be graded on the basis of resistance to passive stretch using Modified Ashworth Scale (MAS)

↓

A 5 point scaling system

↓

Examiner uses passive movt to evaluate resistance to " " due to spasticity

- Modified asworth scale (MAS)

Grade 0 - No ↑ in muscle tone

Grade 1 - Slight ↑ " " "
Minimal resistance at end of ROM

Grade 1+ - Slight ↑ in muscle tone
Minimal resistance throughout the ROM

Grade 2 - More ↑ in muscle tone throughout ROM
but affected part moved

Grade 3 - Considerable ↑ in muscle tone
Passive movt difficult

Grade 4 - Affected parts rigid in flex, ext.

(ii) Reflexes

- Reflexes altered and can vary a/c to stage of recovery

  - In flaccidity - hyporeflexia
  - In spasticity - Hyperreflexia

- Deep tendon reflex active
  - clonus
  - clasp knife response
  - Babinski +ve

(iii) voluantry movt.

- voluantry " are disturbed completely in the affected side
- Two distinct abnormal synergy patterns are described for each extremity
  (i) Flexion synergy
  (ii) Extension "

- There are certain muscles that are not involved in both synergy. So, their activation is difficult
  - These muscles are
    - lattisimus dorsi
    - Teres major
    - Serratus and
    - fingers extensor
    - Ankle everstors

- Flexion synergy components
  - U/E - Scapular retraction / Elevation / Hyperextension
    Sh abd and ext rot
    Elbow flexion
    Forearm supination
    Wrist and finger flexion

L/E - Hip flex, abd, ext rot.
- Knee flex.
- Ankle dorsiflex, inversion
- Toe "

• Extension Synergy component

U/E · Scapular protraction
- Sh. add, int rot
- elbow ext.
- forearm pronation
~~wrist add~~.

L/E - Hip ext, add, Int rot
- Knee ext
- Ankle plantar flex., eversion
- Toe "

(iv) coordination

- Timing and sequencing of muscle impairs the fn. and limits the task performance

- ~~MCA strokes are very common~~
U/

(v) MMT (Muscle strength)

- MCA strokes are very common
↓
UE is more effected than LE
↓
20% Patients fails to regain fn. use of U/E

- MMT of the muscles should be checked and
" " given proper grade

- Postural control and Balance
  - Done by Berg Balance Test
- Gait analysis
  - Circumductory gait test

Goal of PT Management :

- Regaining motor fn.
- " Sensory fn.
- ↓ Pain
- Improve Gait
- " muscle strength and performance
- " ADL
- Prevent other complications
- Improve postural control and balance

PT Management

- Various techniques and approaches are available for the rehab of stroke patients.
- But integrated approach is benificial.

- Positioning
  - Supine - Prone
  - " - Sidelying
  - Sidelying - Prone
  - supine - sitting etc

- Joint flexibility and stability
  - STM
  - Mobilization
  - ROM ex
  - ~~Act~~ Passive movt
  - AROM c̄ terminal stretch
  - Positioning
  - Scapula should be mobilized on thoracic wall c̄ emphasis on upward rotation and protraction.
  - Full elbow extension
  - Edema and tonal change may produce impingement c̄ wrist extension.

    In this case, carpal bones should be mobilized before stretching.
  - Self ROM strategy.
  - Splints are used
  - When sitting in wheelchair, UE can be positioned as:

    Shoulder - 5° abd
    Flexion
    Neutral rotation

    Elbow - 90° flexion
    slightly forward

    Forearm - Pronated

    Hand - Functional resting position

- for plantar flexor spasticity | dorsiflexor weakness.

  • wt shifting activity in modified plantigrade (forward shift stretches plantar flexor)

- stretching

- Active contraction of dorsiflexors.

• Strength

- Sit to stand
- Toe stand
- Stair climbing
- PRE
- Iso kinetic ex.

  • for patients who are very weak (MMT < 3)

    - Gravity eliminating ex.

      ↓

      Powder boards
      Sling
      Suspension
      Aquatic ex

  • for patients $\bar{c}$ MMT = 3

    - Gravity resisted active movt.

      ↓

      Arm lift
      legt "

- for patient c̄ MMT > 3

  - Ex c̄ manual resistance

    ↓

    - free weight
    - Bands
    - Machines

- Resistance training should occur 2-3 times a week.
- 3 sets of 8 - 12 reps per session.
- Resistence training c̄ task oriented activity

  - Sits to stand
  - wall squat
  - Step - ups
  - stair climbing c̄ wt cuffs.

- Rhythmic rotation - slow gentle rotation of limb while progressively moving the limb into lengthening range.

- Hand is kept as wt bearing in Pt side

- Slow rocking movt in sitting
- kneeling
- Quadruped position
- Axial trunk rotation

  ↓

  Side lying

  sitting

  Hook lying

- PNF patterns
- side sitting on hemiparetic side
- Active ex
- cold, massage
- Electrical stimulation
- orthotic device
- Bridging
- Half bridging
- Briding $\bar{c}$ rotation of trunk
- Static balance ex
- Dynamic " "
- Forward bending
- side kicking
- Trunk rotation
- Backward walking